VIT Vidyalankar Institute of Technology
Accredited A+ by NAAC

Version 09/27-09

Consolidated Academic Administration Plan for the Course

OEC 15 Professional Competency Development 2 Sem. V – Program All 2026-27 Odd Semester

Faculty - Prof. (Dr) Rugved V Deolekar

(Cluster Mentor) & Prof. (Dr) Rugved V Deolekar

The academic resources available in VIT –

VMIS (ERP)	V-Refer and V-Live	VIT Library	VAC & MOOC Courses
Institute & Department Vision and Mission	Former IA question papers and solutions (prepared by faculty)	Former IA question papers solutions - hardcopy	Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs
Program Educational Objectives (PEO)	MU end semester examination question papers and solutions (prepared by faculty)	MU end semester exam question paper & solutions - by faculty, hardcopy	
Program Specific Outcome (PSO)	Class notes and Digital Content for the subject (scanned / typed by faculty)	All textbooks, reference books, e -books mentioned in the syllabus & AAP	Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified
Program Outcome (PO)	Comprehensive question bank, EQ, GQ, PPT, Class Test papers	Technical journals and magazines for reference	
Departmental Knowledge Map	Academic Administration Plan & Beyond Syllabus Activity report	VIT library has many resources e. g.: - IEEE, Nimbus, xplore, EBSCO etc.	Watch former lectures captured in LMS at VIT

1.a Course Objectives (Write in detail – as per NBA guidelines)

Cognitive	What do you want students to know?	Students should know the concepts, implementation, analysis, and practical application of major data structures and algorithmic techniques using Java, with emphasis on problem-solving patterns, complexity analysis, competitive coding approaches, and technical interview preparation.
Affective	What do you want students to think / care about?	They should care about efficiency, clarity, and correctness - understanding that a good solution is not just one that works, but one that is logically sound, optimized, and maintainable.
Behavioural	What do you want students to be able to do?	Students should be able to analyze computational problems, select suitable data structures and algorithms, implement efficient Java-based solutions using the Collections Framework, evaluate performance through complexity analysis, and solve coding challenges relevant to competitive programming and technical interviews.

Advice to Students:

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if, what you understand is correct.

The v-Refer Link

https://vrefer.vidyalankar.edu.in/index.php/apps/files/files/921007?dir=/vit%20vrefer/INFT/SEM%20V/2026-27/Verticals/MDC-OE/Subjects/PCD%20%28OEC15%29

Creation of microsite or teams link – (Division wise)

Div A - General | TE INFT A - PCD 2

Div B - General | TE INFT B - PCD 2

Div C - General | TE INFT C - PCD 2

Collaboration Policy:

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the internet, you may use their idea for the solution; provided you acknowledge the source (name and page in the book or the website, if the idea is found on the internet). Even though you are allowed to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible, please ask the teacher.

1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

CO No.	Statements	Related Module/s
CO1	Describe and differentiate core Java Collections like lists, sets, maps, queues, and heaps with respect to their key operations, typical use cases, and trade-offs.	1
CO2	Apply sliding window, prefix sums, hashing, and two-pointers to solve coding problems efficiently.	2
CO3	Implement stack- and queue-based solutions and apply them to expression evaluation, recursion simulation, and other linear-time problem patterns.	3
CO4	Implement linked-list solutions like reverse, rotate, merge sorted lists, and detect cycles and intersections using correct pointer techniques and linear-time approaches.	4
CO5	Develop and analyze tree-based solutions by writing iterative traversals and level order, perform BST insert/delete, and leverage balanced trees, tries, and Huffman coding when they best fit the problem's constraints and efficiency goals.	5
CO6	Model problems as graphs and solve using BFS/DFS, shortest paths, topological ordering, and MST algorithms, with complexity analysis.	6

1.c Course Outcome (CO) Statements mapping with Learning Outcome (LO)

	Mapped to Learning Outcomes

CO1	1.1, 1.3, 3.2, 4.2, 5.2, 6.2
CO2	1.2, 2.1, 2.2, 3.1, 4.1, 4.2, 5.1, 5.2, 6.1, 6.2
CO3	1.3, 3.1, 3.2, 4.1, 5.1, 6.1
CO4	1.2, 2.2, 4.2, 5.2, 6.2
CO5	2.1, 3.1, 3.2, 4.1, 5.1, 6.1
CO6	5.2, 6.1, 6.2

1.d Mapping of COs with POs (As per new detailed Syllabus)

(List of POs is available in V-refer)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	Knowledge A	Analysis	Design	Investigation	Modern Tools	Society	Environment &sustainability	Ethics T	Teamwork C	Communication	Project Mgt	Lifelong learning
CO 1	3/14	1/14	-	-	1/14	-	-	-	-	-	1/14	3/14
CO 2	1/14	4/14	1/14	1/14	-	-	-	-	-	-	1/14	1/14
CO 3	1/14	3/14	3/14	-	1/14	-	-	-	-	-	1/14	1/14
CO 4	1/14	2/14	2/14	1/14	-	-	-	-	-	-	1/14	1/14
CO 5	1/14	2/14	3/14	1/14	-	-	-	-	-	-	1/14	1/14
CO6	1/14	3/14	2/14	1/14	-	-	-	-	-	-	1/14	1/14

1.e Mapping of COs with PSOs (As per new detailed Syllabus)

	PSO 1	PSO 2	PSO 3
CO 1	4/14	1/14	1/14
CO 2	7/14		1/14
CO 3	7/14	1/14	1/14
CO 4	6/14		1/14
CO 5	7/14		1/14
CO6	7/14		1/14

1.f Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course

Verticals	BSC/ESC	Program Courses	Multidisciplinary Courses	Skill Courses	HSSM	Experiential Learning	Liberal Learning
Tick suitable category							

Subject Code	Subject Name	Teaching Scheme			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/Practical	Tutorial	Total
OEC15	Professional Competency Development 2	-	2	-	-	1	-	1

Subject Code	Subject Name	Examination Scheme							
		Theory			Total (Theory)	Practical			Total (Practical)
		ISA	MSE	ESE		ISA	ESE	ORAL	
OEC15	Professional Competency Development 2	-	-	-		50	-	-	50

Subject Code	Subject Name	MSE-1*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
NA				

Subject Code	Subject Name	MSE-2*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
NA				

Subject Code	Subject Name	ESE#		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
NA				

* **Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

Recommended distribution: - 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30 marks from thought provoking

1.g Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course

Divisions	Lecture (Hrs.)	Practical (Hrs.)				Tutorial (Hrs.)			
		Batch 1	Batch 2	Batch 3	Batch 4	Batch 1	Batch 2	Batch 3	Batch 4
A	2 RVD	-	-	-	-	-	-	-	-
B	2 RVD	-	-	-	-	-	-	-	-
C	2 RVD	-	-	-	-	-	-	-	-

1.h Office Hours (Faculty will be available in office in this duration for solving students' query)

Division	Day	Time (at least 1 Hr. / Division)	Venue (Office Room No.)
A	Monday	4.45 to 5.45	IT Dept Staff Room
B	Tuesday	4.45 to 5.45	IT Dept Staff Room
C	Wednesday	4.45 to 5.45	IT Dept Staff Room

Note: Office hours should be after academic hours of students.

2.a Syllabus: Module Wise Teaching Hours and % Weightage in autonomous syllabus Question Paper

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
1	Java Collection Framework	05	20	-	-	PI 1.4.1. PI 1.4.2, PI 2.1.4, PI 2.3.2, PI 3.3.2. PI 1.4.3, PI 2.2.3, PI 3.2.3
Learning Outcome-1.1	*Explain the characteristics and uses of core Java data structures, including arrays, lists, stacks, queues, sets, maps, and priority-based structures.*					
Learning Outcome-1.2	*Differentiate between alternative Java data structures and select the most appropriate one for a given problem.*					
Learning Outcome-1.3	*Use Java Collections Framework classes and their built-in functions to implement efficient problem-specific solutions.*					
2	Advance Concepts and Algorithm Analysis	05	20	-	-	PI 2.1.3, PI 2.2.3, PI 3.2.3,

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
						PI 2.1.2, PI 2.3.2. PI 3.3.1. PI 3.3.2
Learning Outcome-2.1	*Apply problem-solving techniques such as strings and substrings, arrays and hashing, sliding window, and prefix sum to solve coding problems.*					
Learning Outcome-2.2	*Explain and analyze algorithm efficiency using asymptotic analysis, growth functions, recurrence relations, and the Master Method.*					
3	Advance Concepts on Stack, Queues	05	20	-	-	PI 2.1.3, PI 2.2.3, PI 3.2.1, PI 3.2.3, PI 2.1.2, PI 2.2.3, PI 3.2.2
Learning Outcome-3.1	*Apply advanced stack and queue concepts such as monotonic stack, monotonic queue, min-stack, max-stack, deque, priority queue, heap, and stack-queue conversion techniques to solve coding problems.*					
Learning Outcome-3.2	*Explain and implement the use of stack-based techniques in expression conversion and evaluation, recursion simulation, and backtracking applications*					
4	Advance Concepts on Linked List	05	20	-	-	P.I.- 2.1.3, P.I.- 2.2.3, P.I.- 3.2.1, P.I.- 3.2.3, P.I.- 1.4.2, P.I.- 2.1.2, P.I.- 2.1.4, P.I.- 3.3.2
Learning Outcome-4.1	*Apply linked list techniques such as cycle detection, slow-fast pointers, intersection detection, reversal, rotation, merging, palindrome checking, and number representation to solve coding problems.*					
Learning Outcome-4.2	*Explain and compare linked list representations and their applications, including array representation, array vs linked list, polynomial representation, and array of linked lists.*					
5	Advanced Concepts on Trees	05	20	-	-	P.I.- 2.1.3, P.I.- 2.2.3, P.I.- 3.2.1, P.I.- 3.2.3, P.I.- 1.4.2, P.I.- 2.1.2, P.I.- 2.1.4, P.I.- 3.3.2
Learning Outcome-5.1	*Apply tree traversal and binary search tree operations, including iterative inorder/postorder traversal, level order traversal, diameter computation, and deletion in BST, to solve coding problems.*					
Learning Outcome-5.2	*Explain and analyze advanced tree structures and applications, including height-balanced checks, AVL trees, tries, red-black trees, expression trees, and Huffman encoding.*					
6	Advance Concepts on Graphs	05	20	-	-	P.I.- 2.1.3, P.I.- 2.2.3, P.I.- 3.2.1, P.I.- 3.2.3, P.I.- 1.4.2, P.I.- 2.1.2,

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
						P.I.- 2.1.4, P.I.- 3.3.2
Learning Outcome-6.1	*Apply graph traversal and path-finding techniques, including visited-array-based traversal, source-to-target path exploration, shortest path methods, and problem solving on directed and undirected graphs.*					
Learning Outcome-6.2	*Explain and analyze graph representation and advanced graph concepts, including topological sorting, minimum cost spanning tree algorithms, and directed acyclic graphs.*					
	Total	**30**	**100**	**-**	**-**	

Note: - As an attachment Annexure is required for assessment criteria of learning outcomes.

2.b Prerequisite Courses

No.	Semester	Name of the Course	Topic/s
1	4	Professional Competency Development 1	All
2			
3			

2.c Relevance to Future Courses

No.	Semester	Name of the Course
1	6	Professional Competency Development 3
2		
3		

2.d

See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video)

Real Life Scenario	Concept Used
--	

3 Past Results – Division-Wise

Details	Target – DEC 2026	MAY 2025	DEC 2024	MAY 2024

Course Passing % – Average of 2 Divisions	NA			
Marks Obtained by Course Topper (mark/100)				

	Division A		Division B		Division C	
Year	Initials of Teacher	% Result	Initials of Teacher	% Result	Initials of Teacher	% Result
May 2025	NA					
May 2024						
May 2023						

4 All the Learning Resources – Books and E-Resources

4.a List of Textbooks (T – Symbol for Textbooks) to be Referred by Students

Sr. No	Textbook Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Algorithms	S. Dasgupta, C. H. Papadimitriou, and U. V. Vazirani	New York, NY, USA: McGraw-Hill	3.2	1 to 6	Yes
2	Data Structures and Algorithms in Java	M. T. Goodrich, R. Tamassia, and M. H. Goldwasser	Wiley	6	1 to 6	Yes
3	JAVA: The Complete Reference	Herbert Schildt	Oracle Press	13	1 to 6	Yes
4						
5						

4.b List of Reference Books (R – Symbol for Reference Books) to be Referred by Students

Sr. No	Reference Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Data Structures and Algorithm Analysis in Java	M. A. Weiss	Addison-Wesley	3	1 to 6	Yes
2	Algorithms	R. Sedgewick and K. Wayne	Addison-Wesley	4	1 to 6	Yes
3	JAVA Programming- Black Book	Stevan Jolzner	Dreamtech Press	-	1 to 6	-
4						

4.c List of E - Books (E – Symbol for E-Books) to be Referred by Students

Sr. No	E- Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Open Data Structures: An Introduction http://opendatastructures.org/ods-java.pdf	Pat Morin	AU Press	NA	All	Yes
2	Data Structures and Algorithm Analysis in C++ http://people.cs.vt.edu/~shaffer/Book/C++3e20120102.pdf	Clifford A. Shaffer	Dover Publications	NA	All	Yes
3						
4						

4.d **Reading latest / top rated research papers (at least 5 papers)**

Name of Paper	Name of Authors (Background)	Published in		Problem Statement	Available in our Library
		Date	Journal		
NA					

4.e **Based on research paper an identify the current Problem statement**

Problem Statement			Used in			
	Quiz	Assignment	*Mini Project*	Poster Presentation	Test	Any Other
NA						

4.f **Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements**

Name of the Company	To be / Contacted for		
	Student Internship	Student Final Placement	Faculty Internship
NA			

4.g **Identify suitable relevant TOP Guest Speakers from Industry, Example: - (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)**

Name of the Identified Guest Speaker	Designation	Name of the Company
NA		

4.h **Identify relevant technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]**

Name of the Relevant Technical Competition Identified to participate	Organized by	Date of the Event
NA		

4.i **Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication**

University	Name of the Course	Name of Faculty	Type of Collaboration		
			Exchange of Lecture Material	Joint Publication/ Research	Other
NA					

4.j **Module Best Available in – Title of the best resource [from *4.a* to *4.d* in this AAP] and other details as necessary**

Module No.	Title of the Module	Textbook	Mention the Tile					
			Reference Book	E-books	Journal	E-Journal	Available in our Library	V-refer
1	Java Collection Framework	T1	R1	E1	-	-	Yes	-
2	Advance Concepts and Algorithm Analysis	T1	R1	E1	-	-	Yes	-
3	Advance Concepts on Stack, Queue	T2	R2	E1	-	-	Yes	-
4	Advance Concepts on Linked List	T2	R2	E2	-	-	Yes	-
5	Advance Concepts on Trees	T3	R3	E2	-	-	Yes	-
6	Advance Concepts on Graphs	T1	R2	E1	-	-	Yes	-
7	-	-	-	-	-	-	-	-

4.k **Referred to any top-rated university in that subject for content**

University	Name of the Course	Name of Faculty	Date of Delivery of the Course	Remarks
NA				

4.l **Faculty received any certification related to this subject. List of Certifications Identified / Done**

Course	Certifying Agency	No. of Hours	Level of the Course		Certification		Remarks
			Introductory	Advance Skill Development	Done on	Proposed to be on	
Algorithms	Coursera	40		Advance	-	-	-

Specialization							

4.m **Completed subject wise/cluster wise training with cluster mentor.**
List of relevant Refresher Course Identified / Done

Course	Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/University other than MU)	Certification		Remarks
		Done on	Proposed to be on	
Pedagogy	Focus on Placement Aspects	23/06/2026	-	Placement Interview Questions, Mock Interviews
PBL	LeetCode	23/06/2026	-	Relevant leetcode problems to be identified for providing practice to competitive coding
Sub. Content Training	-	-	-	Conduct a mix of Competitive Coding and Technical Interview Questions

4.n **Best Practices Identified and adopted**

No.	Item	Best Practices Identified		
		Univ. 1: IIT Bombay	Univ. 2: MIT	Univ. 3: IIT Delhi
1	Microsite			
2	Video Lectures	√	√	√
3	Assignments			
4	Mini Project			
5	Assessment Metric			
6	Quizzes	√	√	√
7	Labs/ Practical (PBL)	√	√	√
8	Tests			
9	Peer Assessment			
10	Any Other			

4.o **Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos**

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

No.	Websites / Links	Module Nos.
1	http://freevideolectures.com/Course/2279/Data-Structures-And-Algorithms	1 to 6
2	https://www.coursera.org/learn/data-structures/lecture/UdKzQ/stacks	3
3	https://www.youtube.com/watch?v=PGWZUgzDMYI	3, 4

4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

Sr. No.	MOOC Course Link	Course conducted by – Person / University / Institute / Industry	Course Duration	Certificate (Y / N)
1	Data Structures and Algorithms in Java: https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013501567535734784938 7/overview	Infosys Springboard	1 hr 36 Mins	Y

5 Consolidated Course Lesson Plan

	From (date/month/year)	From (date/month/year)	Total Number of Weeks
Semester Duration	06/07/2026	16/10/2026	15

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
1	1	1	Course Introduction, Significance, ISA Evaluation Rubrics		CO1	Understand		--
	2	1	Basic Introduction to Java Collection Framework		CO1	Understand		T3 Chapter #18 Pg 497 - 500
2	3	1	Introduction to various classes and their inbuilt functions from Java Collection Framework		CO1	Understand		T3 Chapter #18 Pg 500 - 521
	4	1	Problem Specific Usages of various classes		CO1	Understand		T3 Chapter #18 Pg 522 - 570

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
3	5	1	Mock Interview on Topics like Java Collection Framework		CO1	Understand		--
	6	2	Quiz 1 - Quiz on Java Collection Framework Questions		CO1	Understand		--
4	7	2	Discussion on Topics- Algorithm Analysis, Growth Functions, Students can also prepare and present		CO2	Apply		T2 Chapter # 4 – Pg 150 – 170
	8	2	Take at least one Leet Code Problems on Strings and Substrings, Arrays and Hashing. Show output tracing. Encourage students to participate.		CO2	Apply		--
5	9	2	Take at least one Leet Code Problems on sliding window approach. Show output tracing. Encourage students to participate.		CO2	Apply		--
	10	2	Quiz 2 - Quiz on Algorithm Analysis and Take Home Test		CO2	Apply		--
6	11	3	Discussion on Topics - Stack Applications, Expression Conversion, Simulation of Recursion		CO3	Apply		T2 Chapter #6 Pg 225 - 250

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
	12	3	Discuss and take at least one leet code problem on Monotonic Stack and Queue, Min Stack and Max Stack		CO3	Apply		--
7	13	3	Discuss and take at least one leet code problem on Stack using Queue, Priority Queue and Heap		CO3	Apply		--
	14	3	Student Presentation 1		CO3	Apply		--
8	15	3	Quiz 3 - Quiz on Stack and Queue Applications and Poster Presentation		CO3	Apply		--
	16	4	Discussion on Topics - Array vs Linked List, Polynomial Representation, Array of Linked List		CO4	Apply		T3 Chapter # Pg 122 - 137
9	17	4	Discuss and take at least one leet code problem on Detecting Cycle in Linked List and Slow Fast Pointers		CO4	Apply		--
	18	4	Discuss and take at least one leet code problem on Reversing, Rotating, Merging of Linked Lists		CO4	Apply		--

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
10	19	4	Mock Interview on Algorithm Analysis, Stack and Queue		CO4	Apply		--
	20	4	Student Presentation 2 and Take Home Test		CO4	Apply		--
11	21	5	Discussion on Topics - AVL Trees, Tries, Red Black Tree and Huffman Encoding		CO5	Analyse		T3 Chapter # 11 Pg 472 - 486
	22	5	Quiz 4 - Quiz on Linked List		CO5	Analyse		--
12	23	5	Discuss and take at least one leet code problem on Iterative Tree Traversals and Level Order Traversals		CO5	Analyse		--
	24	5	Discuss and take at least one leet code problem on Diameter of Binary Tree and Deletion of Node from BST		CO5	Analyse		--
13	25	5	Discussion on Topics - Graph Representation, Topological Sorting, MCST Algorithms and Quiz 5 - Quiz on Trees		CO5	Analyse		T3 Chapter #14 Pg 620 - 672
	26	6	Guest Lecture by Industry Expert		--	--		--

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
14	27	6	Discuss and take at least one leet code problem on Graph Traversals and Visited Array		CO6	Analyse		--
	28	6	Quiz 6 - Quiz on Graphs		CO6	Analyse		--
15	29	6	Discuss and take at least one leet code problem on All Paths, Shortest Path Problems, Directed and Undirected Graphs		CO6	Analyse		--
	30	6	Revision and Doubt Solving		CO6	Analyse		--
16								

6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

Theory (ISA=___) - NA												Total
Class Participation	Assignment -1	Assignment -2	Assignment -3	Assignment -4	Assignment -5	Assignment -6	Assignment -7	Assignment -8	Assignment -9	Assignment -10	Assignment -11	

Practical (ISA=50)												Total
Class Participation	Submission - 1	Submission - 2	Submission - 3	Submission - 4	Submission - 5	Quizzes	Activity	Attendance				
T.Y. B. Tech INFT	04	04	04	04	04	10	10	10				50

Class Participation	MSE-1	MSE-2	ESE	Total
NA				

7 Assignments / Tutorials Details

Assignment/ Tutorial No.	Title of the Assignments / Tutorials	CO Map	Mapping Bloom Taxonomy Level	Assignment/ Tutorials given to Students on	Assignments to be submitted back on
1	NA				
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					

Analysis of Assignment / Tutorial Questions and Related Resources

Assignment / Tutorial No.	Week No.	Type* (√)			Module No.	Based on #			Question Type (√)	
		OT	CS	DTP		Textbook	Reference Book	Other Learning Resource	Real Life Assignments	Thought Provoking
1										
2										
3										

Assignment / Tutorial No.	Week No.	Type* (√)			Module No.	Based on #			Question Type (√)	
		OT	CS	DTP		Textbook	Reference Book	Other Learning Resource	Real Life Assignments	Thought Provoking
4										
5										
6										
7										
8										
9										
10										
11										

* Tick (√) the Type of the Assignment: Online Tools (OT); Collaborative Assignments (CS); Design /Thought provoking (DTP)

Write number for textbook, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

Tests	Test Dates	Module No.	CO Map	MSE Question Paper Pattern	Policy
Student Seminar	Week 7 Week 10	3, 4	CO3, CO4	--	During Lecture
Pop Quiz	Week 3 5, 8, 11, 13, 14	All	CO1, CO2, CO3, CO4, CO5, CO6	--	During Lecture (Best 5 out of 6)
Mock Interviews	Week 3 Week 10	1, 2	CO1, CO2	--	During Lecture (Extra Activity)
Take Home Test	Week 5 Week 10	All	CO1, CO2, CO3, CO4, CO5, CO6	--	Planned twice in a course
Poster Presentation	Week 8	3, 4, 5, 6	CO3, CO4, CO5, CO6	--	Will be floated at the beginning
Class tests / prelims					
Any other test/exams					

*** Failures of IA test (IA1+IA2) shall appear for IA test in the next semester. There is no provision for re-test in the same semester.**

9. Practical Activities

Practical No.	Module No.	Title of the **Experiments**	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	1	Submission No 1	√		JCF	CO1
2	2	Submission No 2	√		Analysis	CO2
3	3	Submission No 3	√		Stack and Queue	CO3
4	4	Submission No 4	√		Linked List	CO4
5	5	Submission No 5	√		Trees	CO5
6	6	Submission No 6	√		Graphs	CO6
7						
8						
9						
10						
11						

10 Uncovering syllabus with different Activities.

No.	Type of the Activity	Activities	Number of beneficiaries	Other Details – guest profile, feedback, mark sheet, report
1	**Experiential learning/Interaction with Outside World**	1- Guest Lectures by Industry Expert		Guest Lecture of Industry Expert by Mr Kiran Hirve, Senior Software Engineer at TSS Consultancy Pvt Ltd
		2- Workshops		
		3- Mini Project		
		4- Industrial Visit		
		5- Any other activity		
2	**Collaborative & Group Activity**	6- Poster Presentation		Students are encouraged to make posters to highlight industry focussed topics
		7- Minute Papers		
		8- Students Seminars		Students will be presenting important concepts and Leet Code Problems explanation.
		9- Students Debates		
		10- Panel Discussion / Mock GD		
		11- Mock Interview		Planned during the course conduction
		12- Any other activity		
3	**Co-Curricular Activity**	13- Informative videos (NPTEL/YouTube /TEDx/ MIT OW/edX)		NPTEL/ YouTube videos on a specific topic will be shown to students on specific topics
		14- Lecture Capture Usage		All lecture recordings are made available to students via MS Teams and Vrefer
		15- Any other activity		

4	**Tests & Assessments**	16- Class Tests/ Weekly Tests		
		17- Pop Quiz		Quizzes are planned during the course conduction
		18- Mobile App Based Quiz		Quizzes are planned during the course conduction
		19- Open Book Test		
		20- Take Home Test		2 Take Home Tests are planned
		21- Any other activity		

11 AAP

No.	Programme	Course	Uploaded on V-refer	Date
1	TE INFT	Professional Competency Development 2	√	03/07/2026

12 Lecture Guide

No.	Programme	Course	Uploaded on V-refer	Date
1	TE INFT	Professional Competency Development 2	√	03/07/2026

*** Do not delete any activity. Give details for planned events. Write 'NA' for activity Not Planned.**

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

Dr Rugved V Deolekar Faculty 1	Prof Amit Aylani Faculty 2	Dr Ganesh Raghtate Faculty 3

External Industry Mentor	External Academic Mentor	Dr Rugved V Deolekar VIT Cluster Mentor	Program HOD

Annexure:

Assessment Criteria of Learning Outcomes:

Learning Outcomes: The Learner will:	**Assessment Criteria:** The Learner can:	**Evaluated under ISA/MSE/ESE/LAB**
LO 1.1: Explain the characteristics and uses of core Java data structures, including arrays, lists, stacks, queues, sets, maps, and priority-based structures. (P.I.- 1.4.1 and 1.4.2) (CO1)	• The student can explain the basic characteristics of arrays, lists, stacks, queues, sets, maps, and priority-based structures. • The student can identify suitable uses of different Java data structures. • The student can distinguish between linear and non-linear data handling structures where relevant. • The student can relate a given problem scenario to an appropriate data structure.	**ISA**
LO 1.2: Differentiate between alternative Java data structures and select the most appropriate one for a given problem. (P.I.- 2.1.4, 2.3.2 and 3.3.2) (CO2)	• The student can compare two or more Java data structures based on their features and usage. • The student can identify differences between alternative structures such as ArrayList and LinkedList, or HashSet and TreeSet. • The student can select an appropriate data structure for a given problem. • The student can justify the selection based on problem requirements and performance considerations.	**ISA**
LO 1.3: Use Java Collections Framework classes and their built-in functions to implement efficient problem-specific solutions. (P.I.- 1.4.3, 2.2.3 and 3.2.3) (CO1)	• The student can use appropriate Java Collections Framework classes in program implementation. • The student can apply built-in methods of lists, sets, maps, queues, and related classes correctly.	**ISA**

	• The student can write Java programs that solve given problems using suitable collections. • The student can produce correct and reasonably efficient solutions using Java collection classes.	
LO 2.1: Apply problem-solving techniques such as strings and substrings, arrays and hashing, sliding window, and prefix sum to solve coding problems. (P.I.- 2.1.3, 2.2.3 and 3.2.3) (CO2)	• The student can identify the appropriate technique for a given coding problem. • The student can solve problems using strings, hashing, sliding window, and prefix sum methods. • The student can implement correct solutions using the selected technique. • The student can demonstrate logical problem-solving steps in coding tasks.	**ISA**
LO 2.2: Explain and analyze algorithm efficiency using asymptotic analysis, growth functions, recurrence relations, and the Master Method. (P.I.- 2.1.2, 2.3.2, 3.3.1 and 3.3.2) (CO2)	• The student can explain the meaning of time complexity and space complexity. • The student can compare growth rates of algorithms using asymptotic notation. • The student can solve or interpret simple recurrence relations. • The student can justify algorithm choice based on efficiency considerations.	**ISA**
LO 3.1: Apply advanced stack and queue concepts such as monotonic stack, monotonic queue, min-stack, max-stack, deque, priority queue, heap, and stack-queue conversion techniques to solve coding problems. (P.I.- 2.1.3, 2.2.3, 3.2.1 and 3.2.3) (CO3)	• The student can identify appropriate stack- or queue-based methods for a given problem. • The student can implement min-stack, max-stack, deque, heap, or priority queue-based solutions. • The student can solve coding problems using monotonic stack or monotonic queue techniques. • The student can implement queue using stacks and stack using queues correctly.	**ISA**
LO 3.2: Explain and implement stack-based techniques for	• The student can explain the role of stack in expression	**ISA**

expression conversion and evaluation, recursion simulation, and backtracking applications. (P.I.- 1.4.2, 2.1.2, 2.2.3 and 3.2.2) (CO3)	processing and recursion simulation. • The student can implement stack-based solutions for expression conversion and evaluation. • The student can demonstrate the use of stack in undo or backtracking operations. • The student can trace the execution of stack-based applications correctly.	
LO 4.1: Apply linked list techniques such as cycle detection, slow-fast pointers, intersection detection, reversal, rotation, merging, palindrome checking, and number representation to solve coding problems. (P.I.- 2.1.3, 2.2.3, 3.2.1 and 3.2.3) (CO4)	• The student can identify suitable linked list techniques for a given problem. • The student can implement linked list operations such as reversal, merging, and rotation correctly. • The student can use slow-fast pointer methods for cycle detection and related problems. • The student can solve standard linked list coding problems with correct logic.	**ISA**
LO 4.2: Explain and compare linked list representations and their applications, including array representation, array vs linked list, polynomial representation, and array of linked lists. (P.I.- 1.4.2, 2.1.2, 2.1.4 and 3.3.2) (CO4)	• The student can explain different forms of linked list representation. • The student can compare arrays and linked lists based on access, insertion, deletion, and storage behavior. • The student can identify suitable applications of linked list representation. • The student can represent simple application structures such as polynomials using linked lists.	**ISA**
LO 5.1: Apply tree traversal and binary search tree operations, including iterative inorder/postorder traversal, level order traversal, diameter computation, and deletion in BST, to solve coding problems. (P.I.- 2.1.3, 2.2.3, 3.2.1 and 3.2.3) (CO5)	• The student can identify suitable tree traversal or BST operations for a given problem. • The student can implement iterative and level order traversals correctly. • The student can solve coding problems involving BST deletion and tree diameter.	**ISA**

	• The student can write logically correct tree-based solutions.	
LO 5.2: Explain and analyze advanced tree structures and applications, including height-balanced checks, AVL trees, tries, red-black trees, expression trees, and Huffman encoding. (P.I.- 1.4.2, 2.1.2, 2.1.4 and 3.3.2) (CO5)	• The student can explain the purpose and structure of AVL trees, tries, red-black trees, expression trees, and Huffman trees. • The student can identify the role of balancing and ordering in tree performance. • The student can compare advanced tree structures based on use case. • The student can analyze suitable applications of these tree-based structures.	**ISA**
LO 6.1: Apply graph traversal and path-finding techniques, including visited-array-based traversal, source-to-target path exploration, shortest path methods, and problem solving on directed and undirected graphs. (P.I.- 2.1.3, 2.2.3, 3.2.1 and 3.2.3) (CO6)	• The student can identify suitable graph traversal or path-finding techniques for a given problem. • The student can implement graph traversal using visited-state logic correctly. • The student can solve directed and undirected graph problems using appropriate methods. • The student can implement source-to-target and shortest path solutions correctly.	**ISA**
LO 6.2: Explain and analyze graph representation and advanced graph concepts, including topological sorting, minimum cost spanning tree algorithms, and directed acyclic graphs. (P.I.- 1.4.2, 2.1.2, 2.1.4 and 3.3.2) (CO6)	• The student can explain common graph representations and their uses. • The student can distinguish among DAGs, topological sorting, and spanning tree concepts. • The student can identify suitable graph algorithms for different applications. • The student can analyze graph problems using structural and algorithmic reasoning.	**ISA**